Element B

Stakeholders need a design to let disabled kids be able to participate and have an easier time participating. The designs need to be able to work, be easy to use, easy to understand, and easy to transport. The design should make it easier to play hockey.

1. *Adaptive floor hockey equipment - wheelchair hockey stick holder*. YouTube. (2020, November 10). https://youtu.be/do-zZAKmh4c?si=w3eur7nfTIZtqRod

Design is used by a physical therapist or PE teacher. He is using his feet in the video to move the wheels around, but the design can still work with a motorized and regular wheelchair. There's PVC pipes connected to the wheelchair along with one in the middle holding the hockey stick up right in front of the teacher.

The strength of this idea is that the stick will be in front of them at all times and easy to move, so they can focus on moving the wheels. The shortcoming for this design may be the fact that the hockey stick may drag on the floor. The stick may break and it could be hard to reattach.

2. *Wheelchair-mounted hockey stick*. https://www.canassist.ca. (n.d.). https://www.canassist.ca/EN/main/programs/our-technologies/current-technologies/wheelchair-mounted-hockey-stick.html

This guy who made this design for this boy most likely works in physical therapy.
The boy doesn't have to use his hands or arms for the design. The stick consists of a design that has a hockey stick with a cup up plastic cup that makes it easier to catch the ball. It's magnetically attached to the wheelchair. It has easy set up, removes easy, and folds easy for storage and transportation. Strength easy to attach, remove, and transport. The magnetic attachment is good. Easy to hold the ball.

3. *Adapted Hockey stick*. YouTube. (2018, November 20). https://youtu.be/D6cSDObd0X8?si=iCuRagTGdKZ_xCvK

This physical therapist built this wheelchair design that involves hockey sticks. It works by having a person come from behind and push the wheelchair. The hockey sticks are held on the wheelchair with ziplocks, and are held together at the bottom with a pool noodle. It doesn't really meet the stakeholders needs because the kids won't be able to move it around themselves.

4. This is a more basic design that doesn't really involve the engineering that the others do. This is more like just the concept of wheelchair hockey itself.

Sled hockey,a game invented in Sweden, uses a design that is essentially a sled that sits on top of skate blades. The players sit on top of it and hold two hockey sticks instead of one. The hockey sticks have metal picks so that the players can propel themselves forward. It could be less safe for students with low gross motor skills because they need to have enough strength to push themselves forward.

Smith, Meghan. "A Boston Nonprofit Brings Volt Hockey to Us, 'A Perfect Sport' for People with Physical Disabilities." *GBH*, GBH News, 7 Aug. 2023, www.wgbh.org/news/local/2022-05-20/a-boston-nonprofit-brings-volt-hockey-to-us-a-perfect-sport-for-people-with-physical-disabilities.

The Boston Self Help Center, a non profit organization, has introduced a solution to allow people with disabilities to play a new version of hockey called vault hockey.

They developed specially made wooden electric wheelchairs with plastic hockey stick paddles on the end and it is used on the hardwood floor. The wheel chair is able to go up to 10 mph and turns easily. It is operated by hand with a joystick, this allows people with limited upper body mobility to participate. The disadvantage of this design is that it is not very simple to assemble, it is a very complex design.

“Powerchair Hockey.” Power Hockey the Sport, www.topendsports.com/sport/list/hockey-power.htm. Accessed 15 Mar. 2024.

Powerchair hockey is played on a hard floor with an electric wheelchair. Players are able to quickly maneuver their wheelchair and stick to make it competitive. They use a plastic ball instead of a puck and a plastic stick. Disadvantage is that electric powered wheelchairs are too fast paced for kids which could become dangerous for younger users. If the kid doesn’t have much hand mobility it’ll be very difficult for them to play this way since they have nothing helping them in terms of the stick.

Element C

All design requirements are related to prior research
The design requirements are:

1. It has to help the kids socialize with other children. low priority
2. The attempt has to be feasible. High priority
3. It has to be easy for the teachers and kids to be able to use it and figure out how it works. High priority
4. It has to actually help kids and work properly. high priority
5. The design has to be safe. high priority
6. The design should be able to incorporate and hello the needs of the special ed kids low priority

4 of the 6 design requirements listed are identified as high priority.
The design requirements would be highly likely to lead to a tangible and viable problem solution because these requirements are pretty basic and can be achieved quite easily.
There is evidence that requirements represent the needs of, and have been validated by, 100% of the primary stakeholder groups by these requirements being pretty basic and most of them being high priority means that the design we make along with the requirements we’ve given should easily be able to meet the stakeholders needs.